

Rigid Analytic Geometry Week 8 Solution

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Problem 1

Since M is bounded, for any $U \in \mathfrak{Q}_C(A)$ a neighbourhood of 0, there is another neighbourhood of 0, $V \in \mathfrak{Q}_C(B)$ such that

$$VM \subseteq U.$$

Using that f is open we see $f(U), f(V) \in \mathfrak{Q}_C(B)$ $0 = f(0) \in f(U), f(V)$, and

$$f(V)f(M) \subseteq f(U).$$

For any $U' \in \mathfrak{Q}_C(B)$, $f^{-1}(U')$ is open thus if we take $U \in \mathfrak{Q}_C(X)$ such that $U \subseteq f^{-1}(U')$, we have

$$f(U) \subseteq U'.$$

Thus this proves the statement.

Problem 2

For any open set $U \subseteq A$, the saturation of U is $U + I$ which is a union of open sets thus open. Therefore the quotient map is open, by the previous problem, the statement follows.

Problem 3

Clearly, for any $a \in (0, \infty)$, there is $n \in \mathbb{N}$ such that $r^n < a$. Thus Y is topologically nilpotent. For any $a \in (0, \infty)$, then take $x \in K$ such that $|tx| > a$ for some $t \in Y$, then $a\varepsilon$ will not sit in a ball of radius a . This element is contained in any power of $X \cup Y$. Thus it is not topologically nilpotent.

Problem 4

Since horizontal specialization is defined as a restriction and setting other points 0, the inequality on A^+ is preserved.

Problem 5

If $c\Gamma_x \neq \Gamma$, then $x|_{c\Gamma}$ is different from x which is a specialization of x .